

OpenShift / Kubernetes S/4HANA Development Cluster Requirements

Platform: RHEL 9 (All Nodes)

Purpose: Development / POC (Not Production)

1. Overview

This document defines the minimum and recommended hardware and cluster layout required to run containerized SAP S/4HANA development systems on OpenShift using RHEL 9 nodes only.

2. Control Plane Requirements (Master Nodes)

Minimum (Non-Production / Dev)

- 1 × Master Node (Single Master)
- 4 vCPU
- 16–32 GB RAM
- 120 GB SSD
- RHEL 9
- Suitable for dev/POC
- API downtime possible during master maintenance

Recommended (More Stability)

- 2 × Master Nodes
- 4–8 vCPU each
- 32 GB RAM each
- 120–200 GB SSD each
- RHEL 9
- Reduces downtime risks during updates

Reasoning:

1 master = cheapest and simplest.

2 masters = better stability but not full HA.

3 masters = only needed for production, not required here.

3. Worker Node Pools

Pool 1 — S/4HANA High-Memory Nodes

Dedicated for S/4HANA pods only.

Minimum (Cheapest)

- 1 × Worker Node (S/4HANA Dedicated)
- 16 vCPU
- 128 GB RAM
- 500GB–1TB NVMe
- RHEL 9

Recommended

- 2 × Worker Nodes (S/4HANA Dedicated)
- 24–32 vCPU
- 192–256 GB RAM
- 1 TB NVMe
- RHEL 9

Pool 2 — Utility / General Workload Nodes

Runs everything that is not S/4HANA:

- NGINX Reverse Proxy
- Monitoring (Grafana / Prometheus)
- Logging / Backup tools
- CI/CD agents
- Networking workloads

Minimum (Cheapest)

- 1 × Utility Node
- 8 vCPU
- 32 GB RAM
- 200 GB SSD
- RHEL 9

Recommended

- 2 × Utility Nodes
- 8–16 vCPU
- 32–64 GB RAM
- 200–300 GB SSD
- RHEL 9

4. Full Hardware Matrix

Node Type | Count | vCPU | RAM | Disk | OS

Master (Min) | 1 | 4 | 16–32GB | 120GB SSD | RHEL9

Master (Rec.) | 2 | 4–8 | 32GB | 120–200GB | RHEL9

S/4HANA Node (Min) | 1 | 16 | 128GB | 500GB–1TB NVMe | RHEL9

S/4HANA Node (Rec.) | 2 | 24–32 | 192–256GB | 1TB NVMe | RHEL9

Utility Node (Min) | 1 | 8 | 32GB | 200GB SSD | RHEL9

Utility Node (Rec.) | 2 | 8–16 | 32–64GB | 200–300GB SSD | RHEL9

5. Minimal Cluster Options

Option A — Cheapest (2 Nodes)

- Node 1: Master + Utility (8 vCPU, 32GB RAM)
- Node 2: S/4HANA Node (16 vCPU, 128GB RAM)

→ Best for very low budget POCs.

Option B — Recommended (3 Nodes)

- 1 Master
- 1 S/4HANA High-Memory Node
- 1 Utility Node

→ Balanced isolation and cost.

Option C — High-Performance (5 Nodes)

- 2 Masters
- 2 S/4HANA High-Memory Nodes
- 1 Utility Node

→ Best for multiple dev systems / N+M sandbox.

6. Software Requirements

- RHEL 9 (All Nodes)
- OpenShift 4.x
- CRI-O runtime
- NGINX reverse proxy container
- Storage:
 - NVMe local disks or
 - ODF / Ceph for dynamic PVCs
- Load balancer (MetalLB or cloud LB)

7. Networking Requirements

- Pod CIDR: /16
- Service CIDR: /16
- SAP required ports (3200–3299, 44300, 3300, 50000)
- NGINX reverse proxy → forwards SAPGUI ports

- NodePorts or LoadBalancers for external access

8. Development Cluster Workflow

1. Deploy OpenShift with node pools.
2. Build/pull S/4HANA container image.
3. Create YAML for each S/4HANA pod.
4. Configure NodePort/LoadBalancer.
5. Deploy NGINX reverse proxy.
6. Validate SAPGUI/Fiori login.
7. Scale by adding pods using YAML duplication.

9. Jira Breakdown

Epic: S/4HANA Containerized Development Platform

Feature 1: Provision RHEL9 Infrastructure

- Story 1.1: Deploy master node(s)
- Story 1.2: Deploy S/4HANA worker nodes
- Story 1.3: Deploy utility nodes

Feature 2: Install OpenShift

- Story 2.1: Base OpenShift installation
- Story 2.2: Node labeling
- Story 2.3: Cluster validation

Feature 3: Deploy S/4HANA Runtime

- Story 3.1: Configure registry
- Story 3.2: Deploy S/4HANA pods
- Story 3.3: Create SAP services

Feature 4: Networking

- Story 4.1: Deploy NGINX reverse proxy
- Story 4.2: Configure SAP ports
- Story 4.3: Validate SAP access

Feature 5: Observability

- Story 5.1: Deploy monitoring stack
- Story 5.2: Deploy logging stack

10. Final Notes

This document reflects the exact requirements for running SAP S/4HANA containers on OpenShift using RHEL9 nodes in a development or POC environment. It balances cost, scalability, and node role clarity.